LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need for simple oxygen system and pressurized cabin effect of lack of o2, Types of 02 regulators
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need for simple oxygen system and pressurized cabin effect of lack of o2, Types of 02 regulators. Fire detection and protection system:
5. Specific Objectives	23.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of demand regulator operation; Safety: Do not breathe directly from regulator; Instructor supervision required
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft oxygen system?
(b) How does cabin pressure change with altitude?
(c) What are the symptoms of hypoxia in crew members?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Physiological effects of hypoxia • Need for simple oxygen systems • Pressurized cabin effects • Oxygen system types overview	L & Ds	PPT, IB	15 min
(b)	• Working principle of demand regulators • Continuous flow regulator operation • Comparison of regulator types • Safety features in regulators	L & Ds	PPT, PDF	20 min
(c)	• Maintenance requirements for regulators • Troubleshooting common regulator faults • Regulator selection criteria • Integration with fire protection systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between demand and continuous flow regulators.
(b) Describe the effect of lack of oxygen at high altitudes.
(c) List three safety features found in modern oxygen regulators.

12. Home Assignment and Follow Up	Prepare a brief note on how the oxygen system requirements change for LP 02 (Fire detection and protection system) integration.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of fire zone classification of a/c, detection system requirements, types of detection system, types of detectors, extinguishing agents and systems	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min